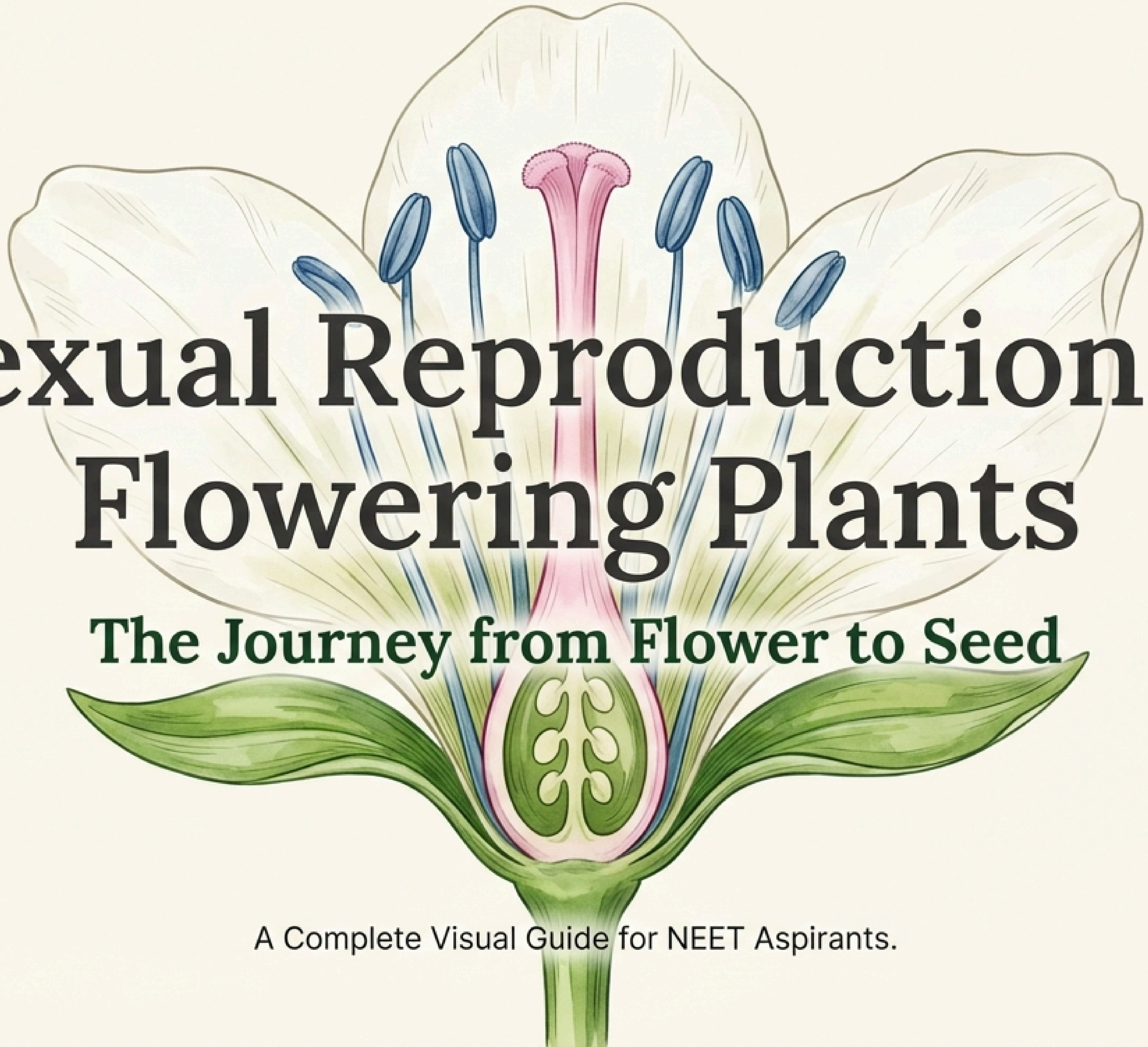


Sexual Reproduction in Flowering Plants

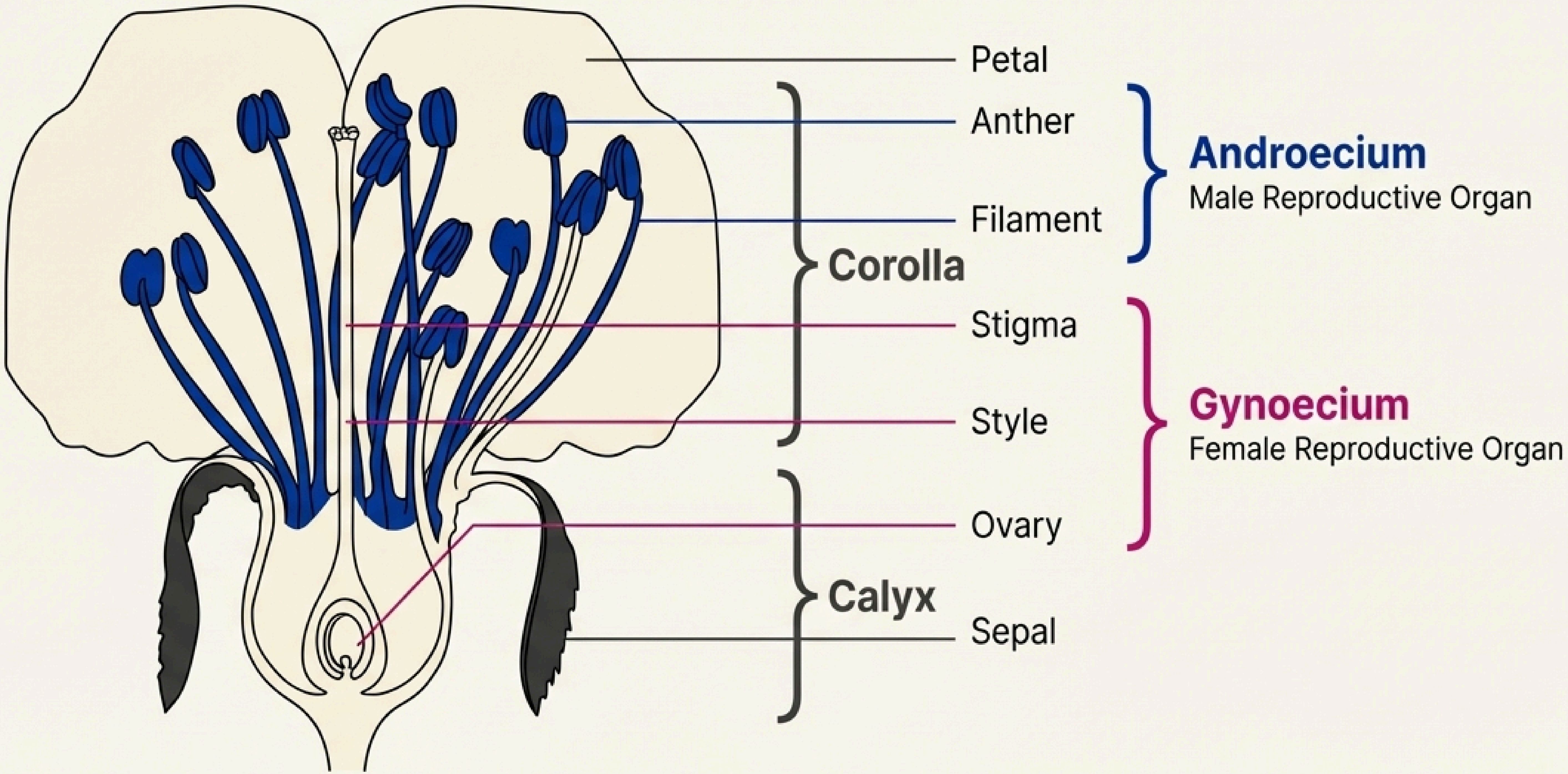
The Journey from Flower to Seed

A Complete Visual Guide for NEET Aspirants.

SOHAN KUMAR

The Flower: An Angiosperm's Fascinating Reproductive Organ

Flowers are the morphological and embryological marvels and the sites of sexual reproduction in angiosperms. All flowering plants show sexual reproduction, employing an amazing range of adaptations to form fruits and seeds.



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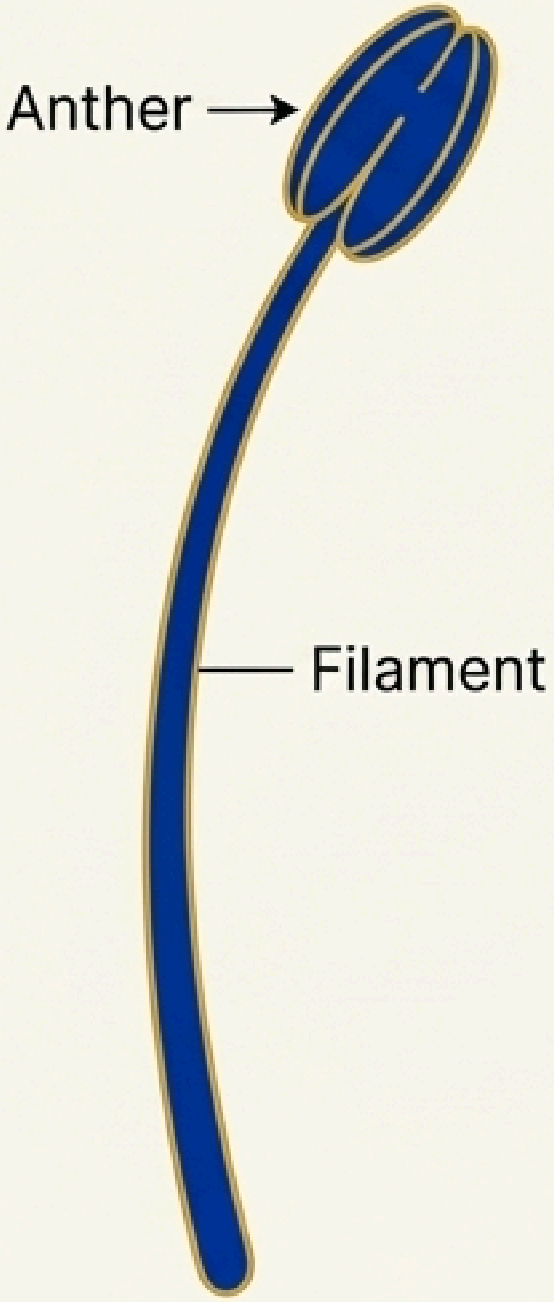
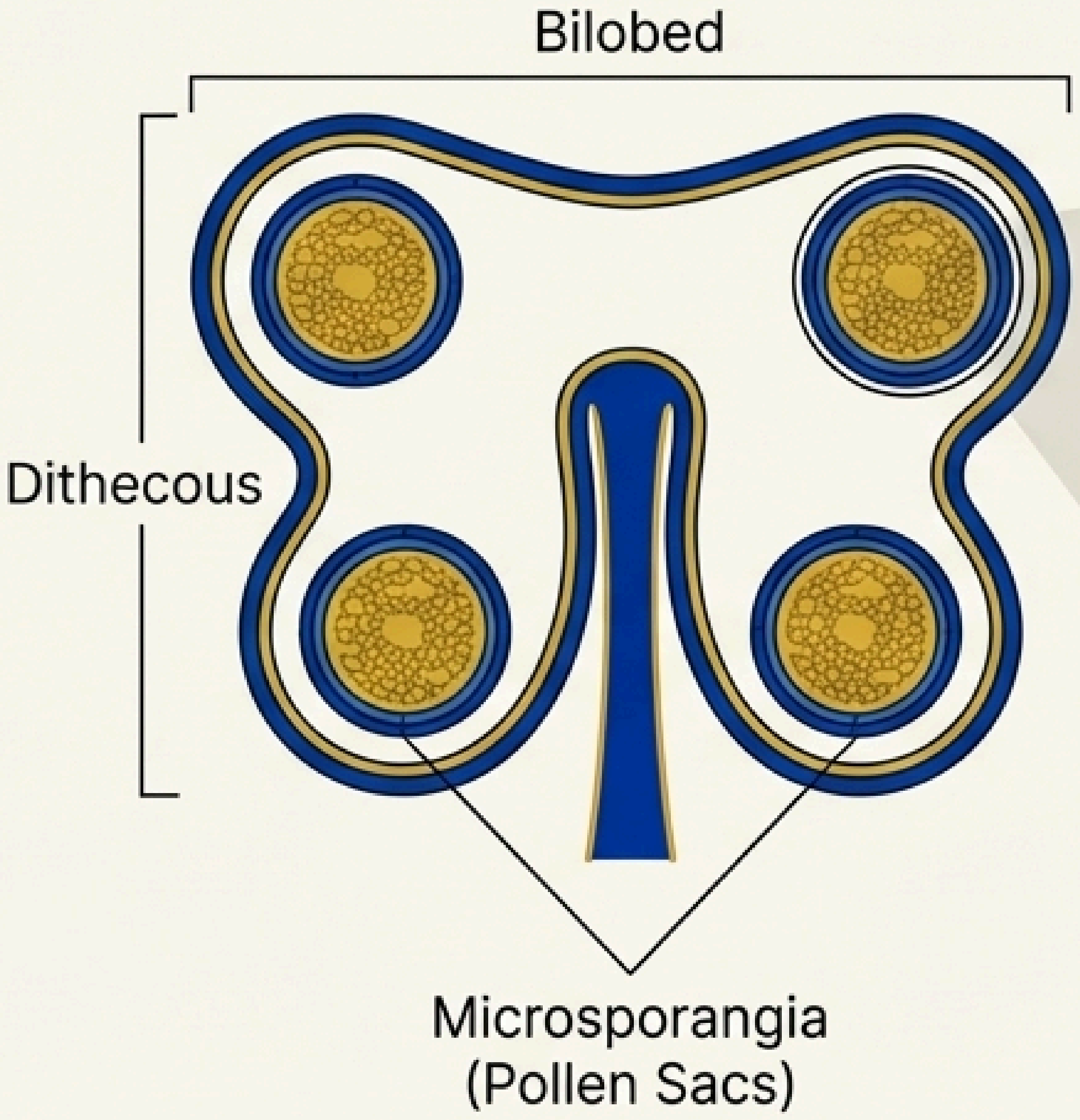
Developing the Male Gametophyte: Inside the Anther

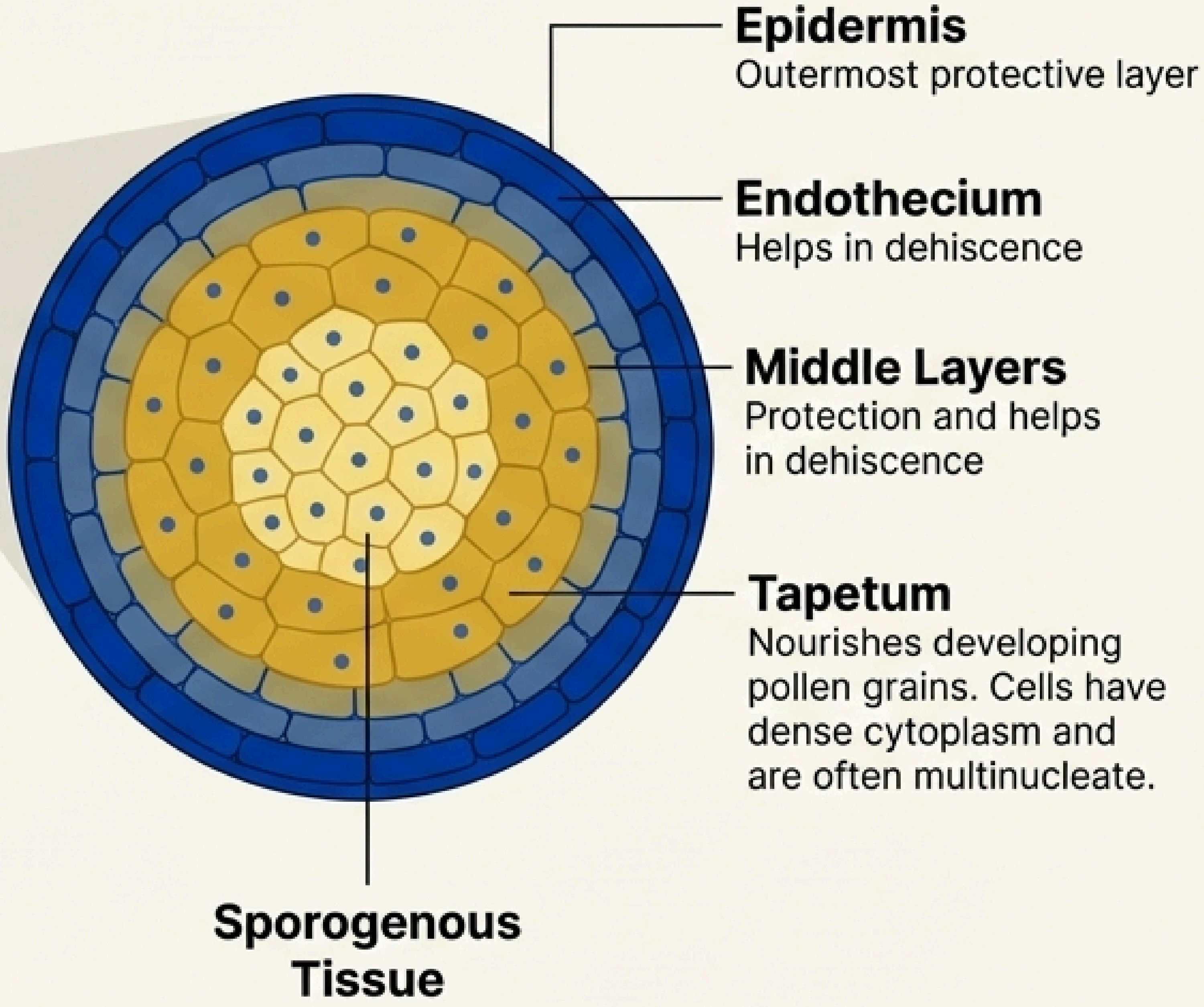
The Stamen



T.S. of an Anther



The Microsporangium Wall



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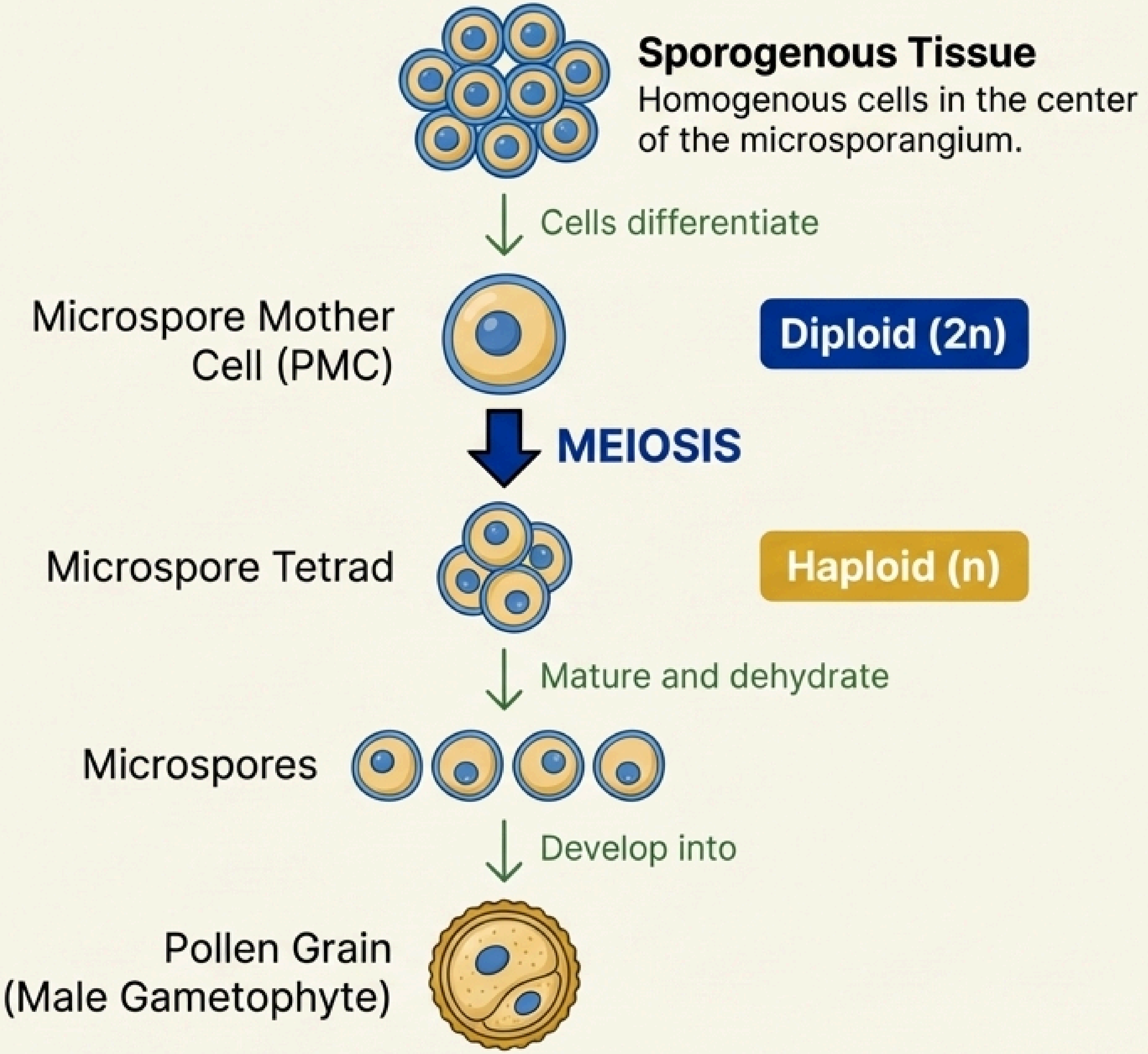
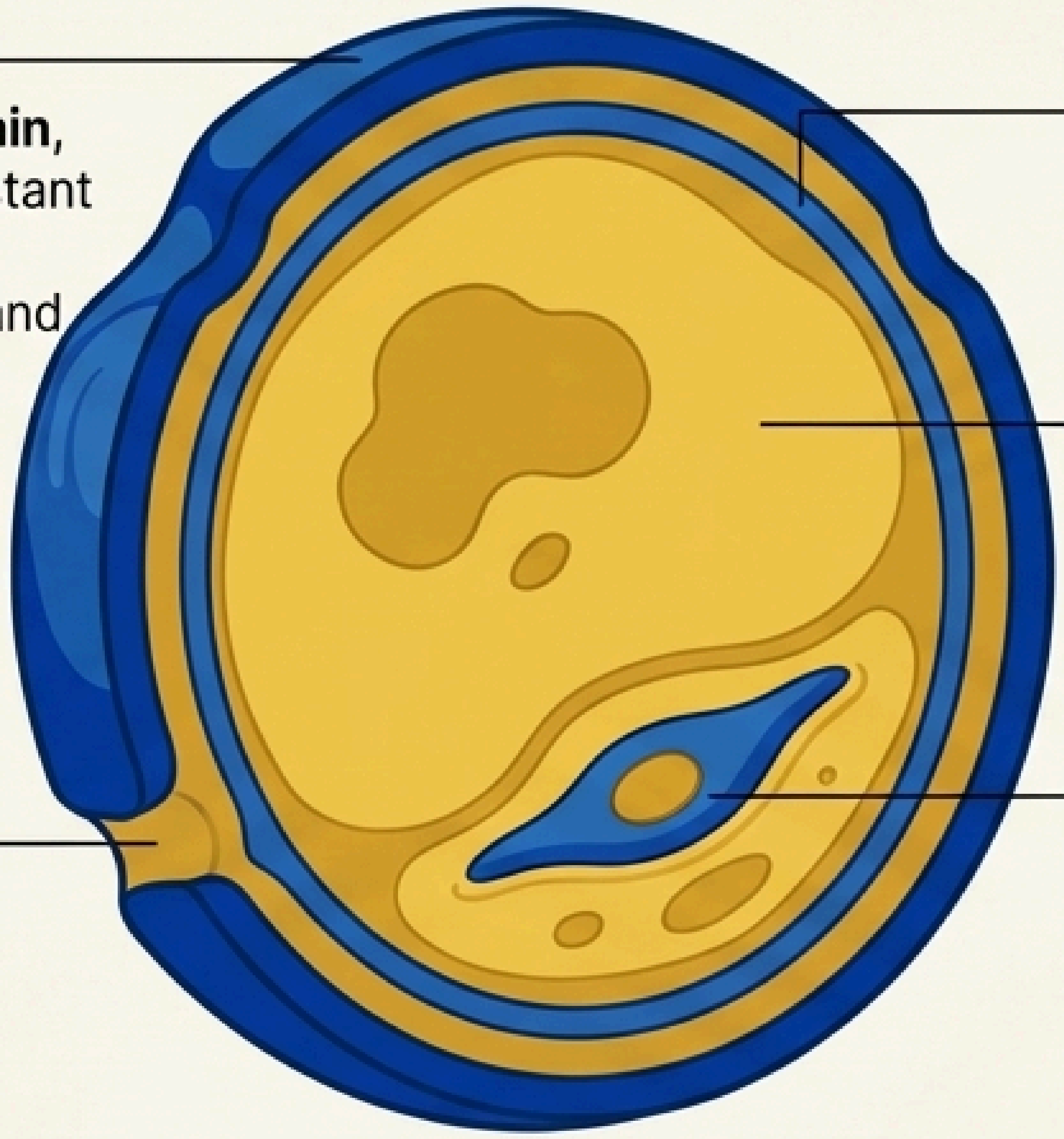
Microsporogenesis: From Mother Cell to Pollen Grain



★ Each cell of the sporogenous tissue is a potential PMC and is capable of giving rise to a microspore tetrad. The process of forming microspores from a PMC through meiosis is called **microsporogenesis**.

The Pollen Grain: A Resilient Male Gametophyte



Exine
Made of **Sporopollenin**, one of the most resistant organic materials known. It can withstand high temperatures, strong acids, and alkali.
This is why pollen is well-preserved in fossils.

Germ Pore

Intine
Made up of **cellulose and pectin.**

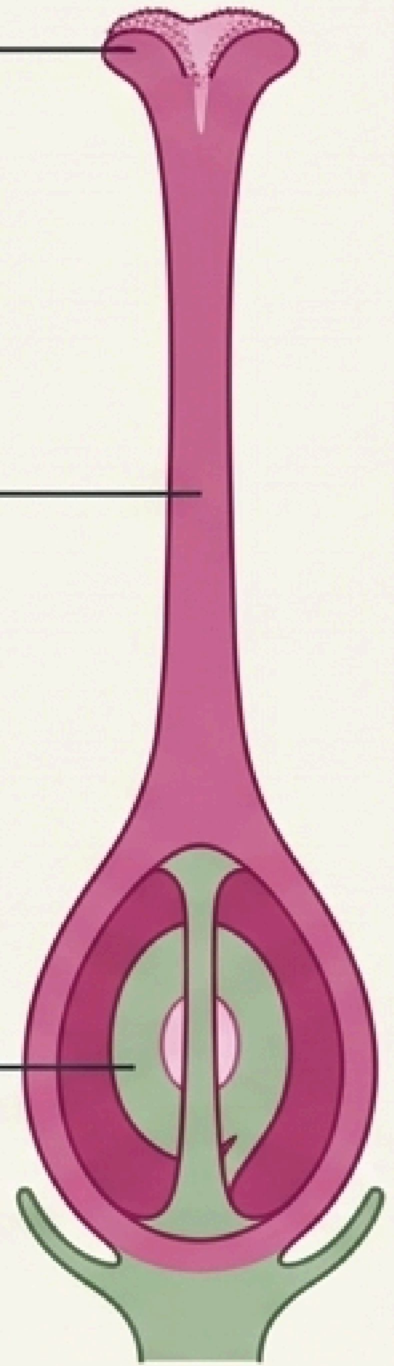
Vegetative Cell
Has abundant food reserve and a large, irregularly shaped nucleus.

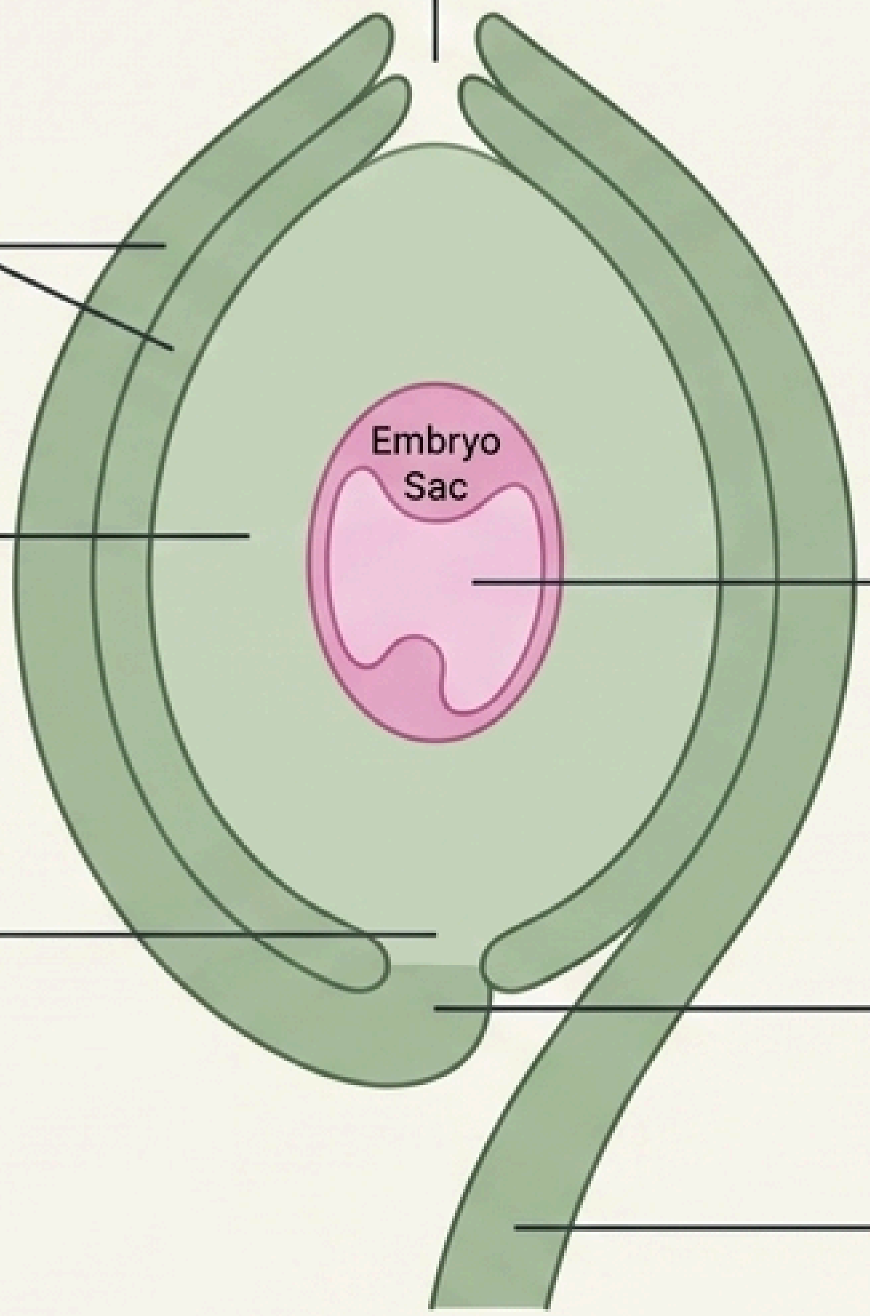
Generative Cell
Dense cytoplasm and a nucleus.

High-Yield Fact

In over 60% of angiosperms, pollen grains are shed at this 2-celled stage. In the remaining species, the generative cell divides mitotically to form two male gametes before shedding (3-celled stage).

Developing the Female Gametophyte: The Pistil and Ovule





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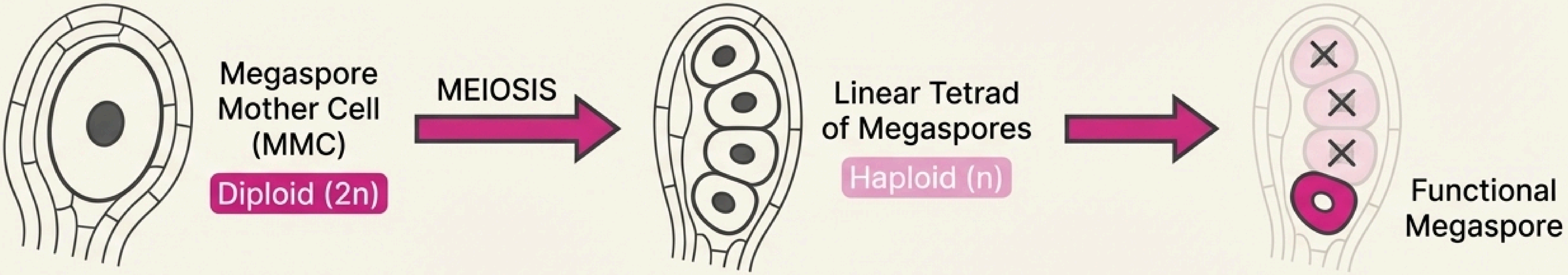
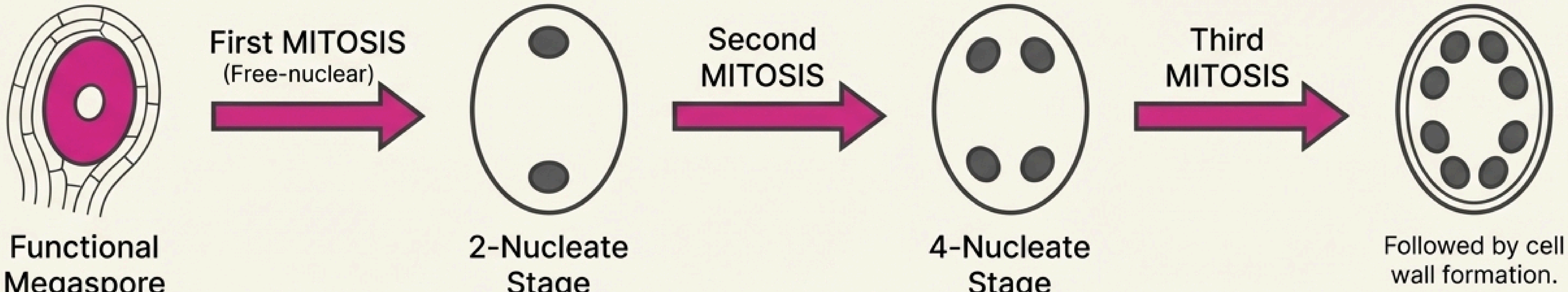
Megasporogenesis & Embryo Sac Development

Part 1: Megasporogenesis - Forming the Megaspore



Part 2: Monosporic Development - Forming the Embryo Sac



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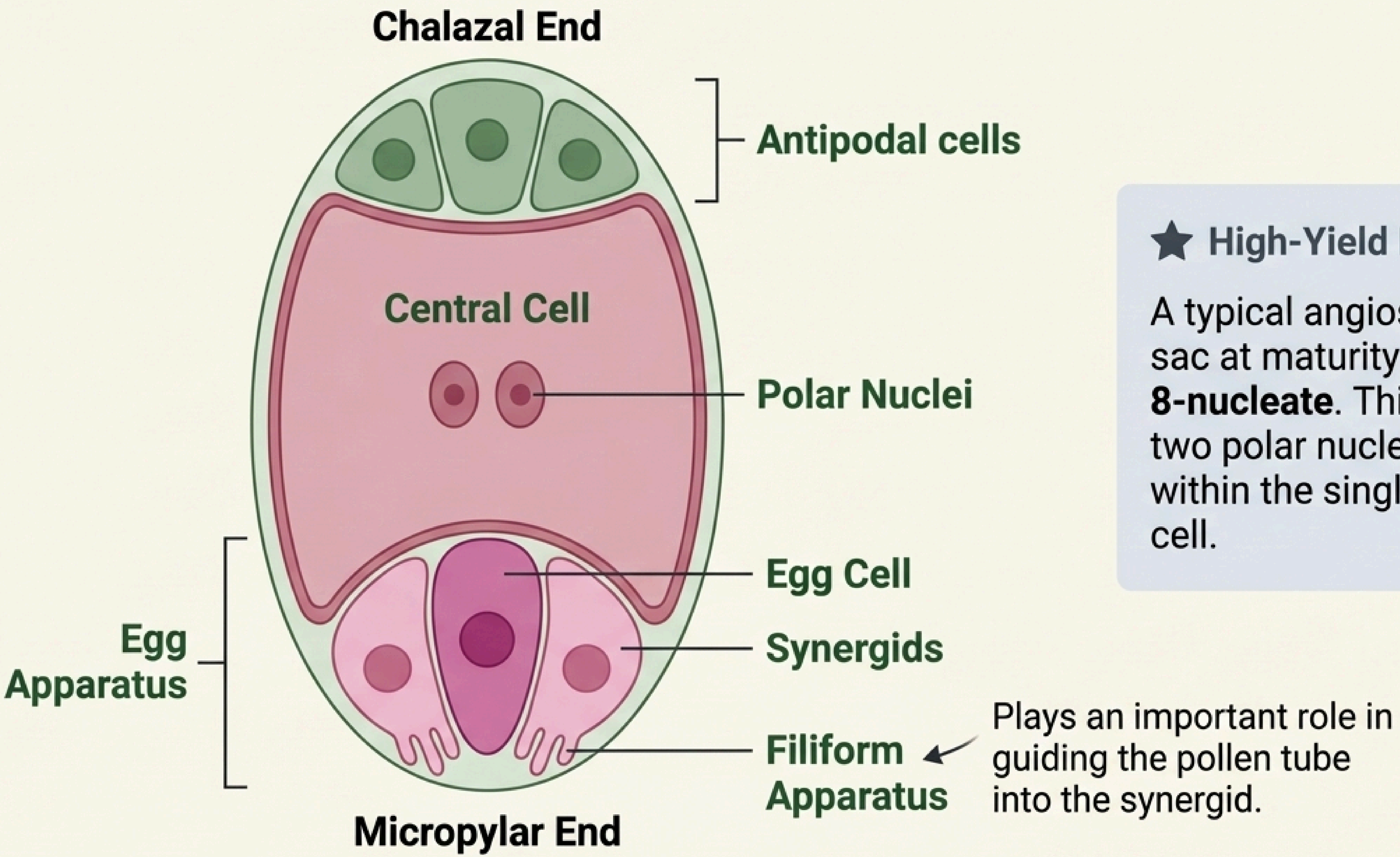
The Mature Embryo Sac: 7-celled, 8-nucleate



★ **High-Yield Fact**

A typical angiosperm embryo sac at maturity is **7-celled and 8-nucleate**. This is because the two polar nuclei are located within the single large central cell.

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Pollination: Bringing Gametes Together

Pollination is the transfer of pollen grains from the anther to the stigma of a pistil.

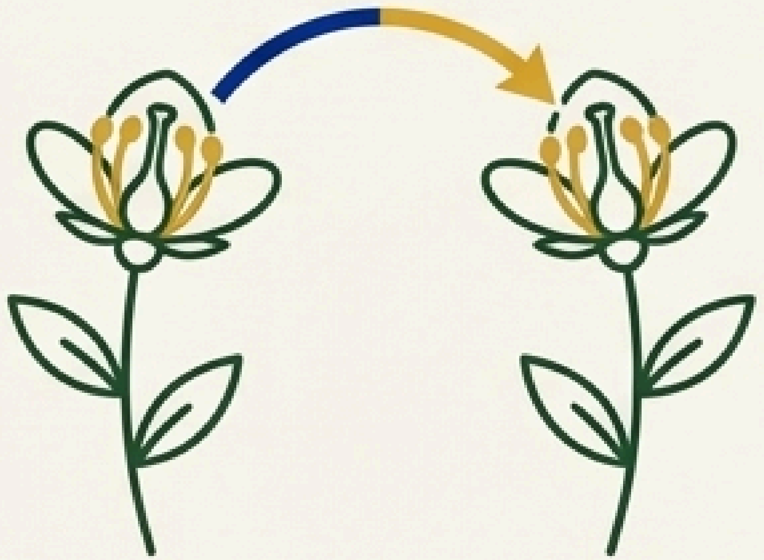
Autogamy (Self-Pollination)

Pollination is achieved within the same flower.

Geitonogamy

Transfer of pollen to another flower of the same plant. Functionally, this is cross-pollination, but genetically it is similar to autogamy.

Xenogamy (Cross-Pollination)



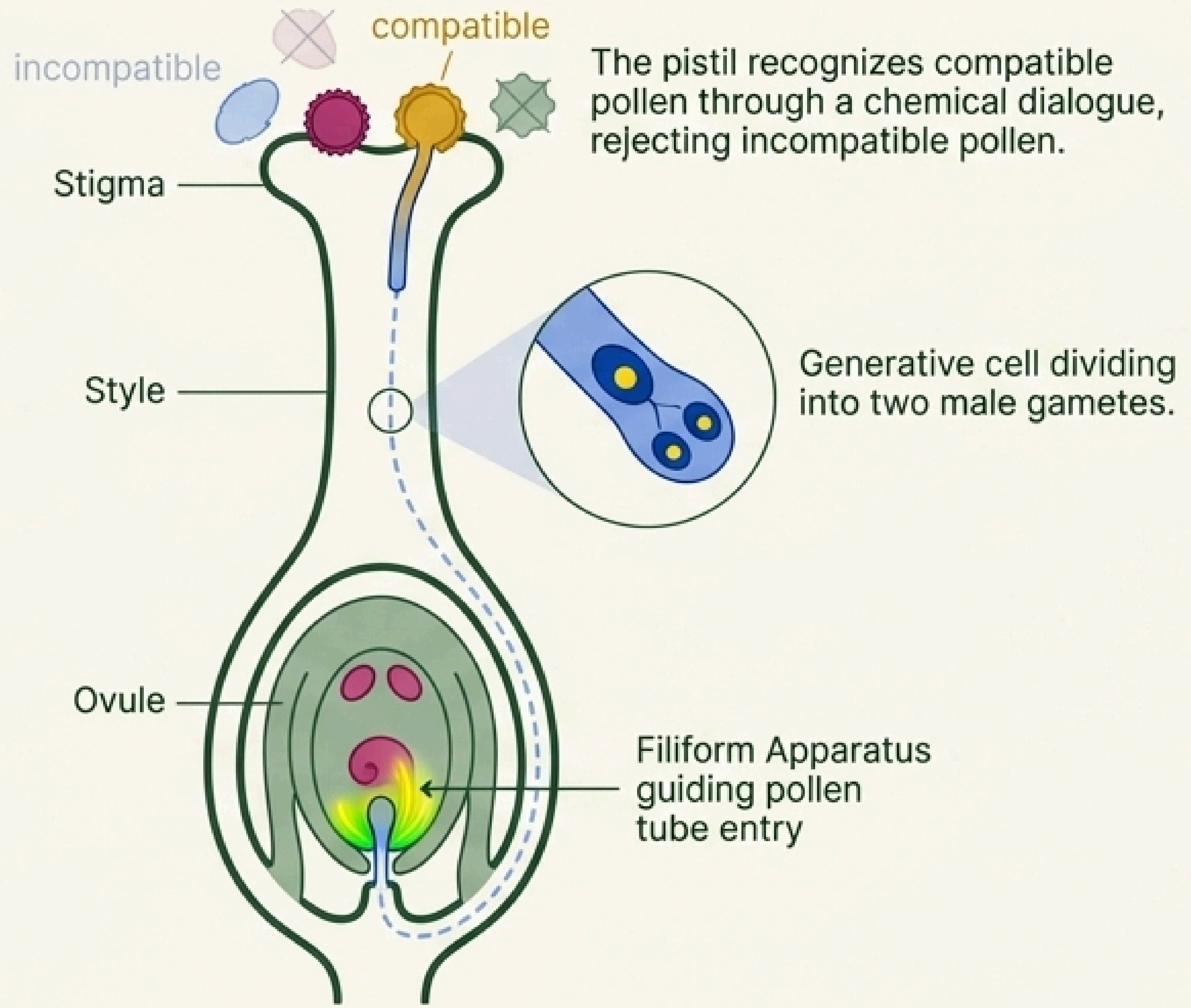
Transfer of pollen to the stigma of a different plant. This is the only type of pollination that brings genetically different pollen to the stigma.

Agents of Pollination

Abiotic Agents: 

Biotic Agents: 

Pollen-Pistil Interaction: The Chemical Dialogue

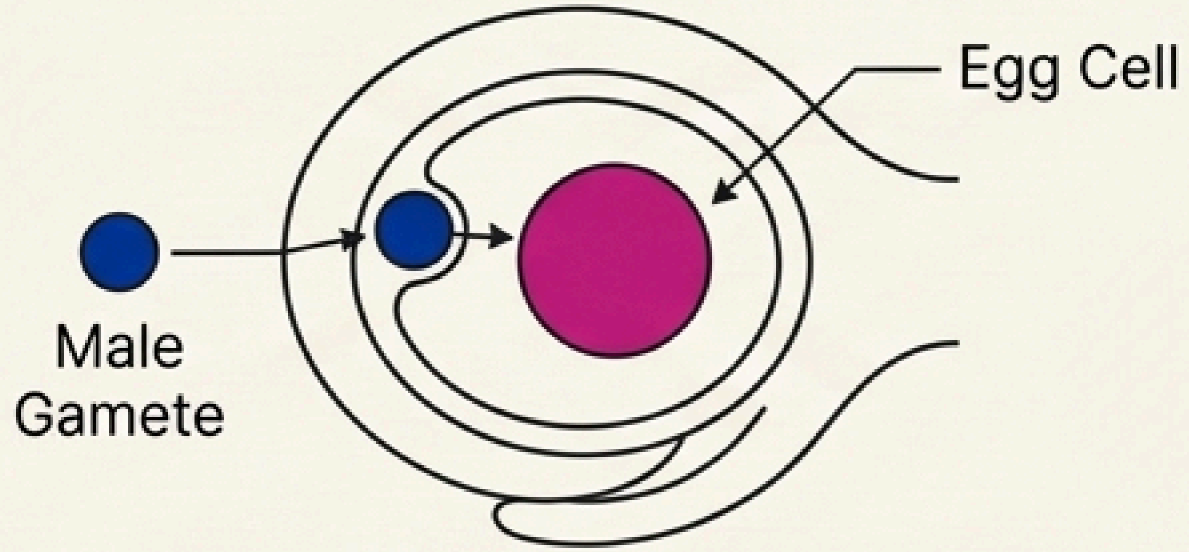


All events—from pollen deposition on the stigma until the pollen tube enters the ovule—are together referred to as pollen-pistil interaction.

Double Fertilization: The Signature Event of Angiosperms

After entering the synergid, the pollen tube releases two male gametes. Two types of fusions occur in the embryo sac, a phenomenon termed **Double Fertilization.**

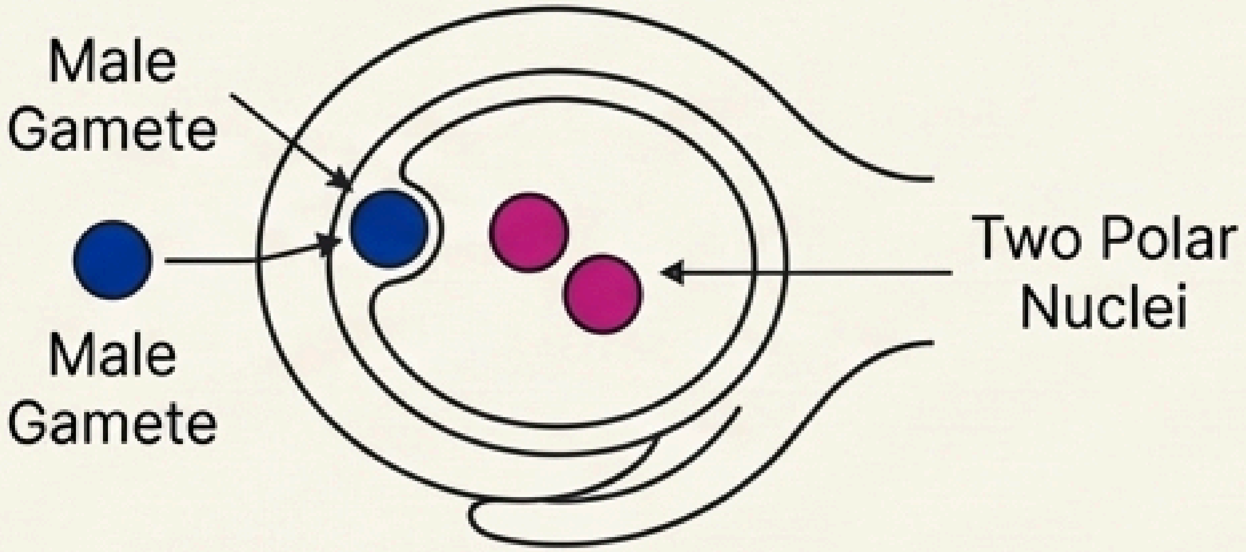
Syngamy (True Fertilization)



Male Gamete (n) + Egg Cell (n) → Zygote (2n)

↳ Develops into the **Embryo**.

Triple Fusion



Male Gamete (n) + Two Polar Nuclei (n + n) → Primary Endosperm Nucleus (PEN) (3n)

↳ Develops into the **Endosperm** (nutritive tissue).

Double Fertilization = Syngamy + Triple Fusion

High-Yield Fact ★ Double fertilization is an event unique to flowering plants.

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Post-Fertilization: The Transformation into Seed & Fruit

Following double fertilization, the ovule matures into a seed, and the ovary develops into a fruit. This table summarizes the fate of the key floral parts.

Floral Part (Before Fertilization)	Develops Into (After Fertilization)
Zygote →	**Embryo**
Primary Endosperm Nucleus (PEN) →	**Endosperm** (nutritive tissue, 3n)
Ovule →	**Seed**
Integuments of Ovule →	**Seed Coat** (Testa & Tegmen)
Ovary →	**Fruit**
Ovary Wall →	**Pericarp** (Fruit Wall)
Synergids & Antipodals →	*Degenerate*

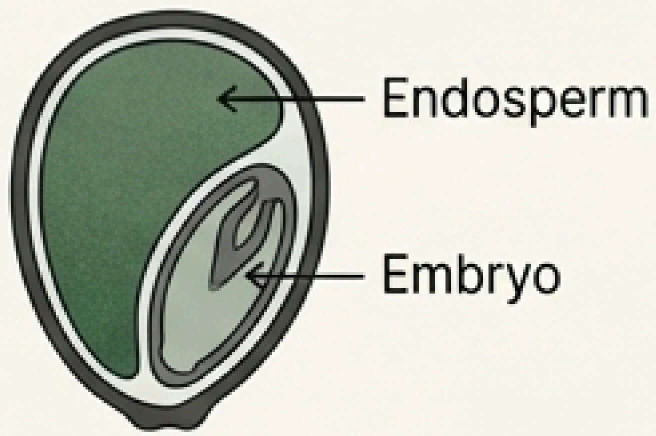
Embryo & Endosperm: The Seed's Core Components

The Endosperm

Function: Provides nutrition to the developing embryo. Its development precedes embryo development to ensure an assured food supply.

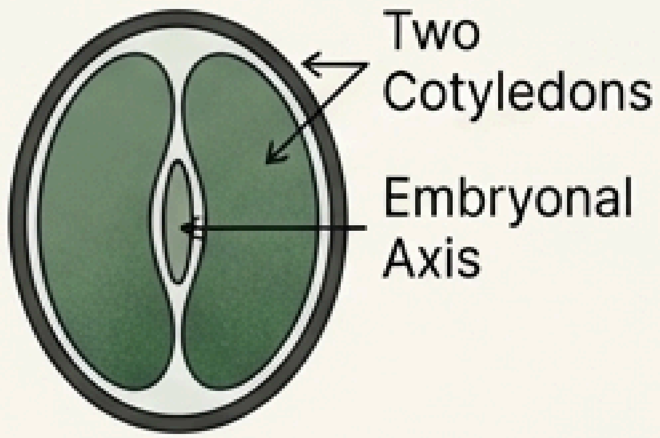
Albuminous (Endospermic) Seeds

Retain endosperm at maturity (e.g., Wheat, Maize, Castor).



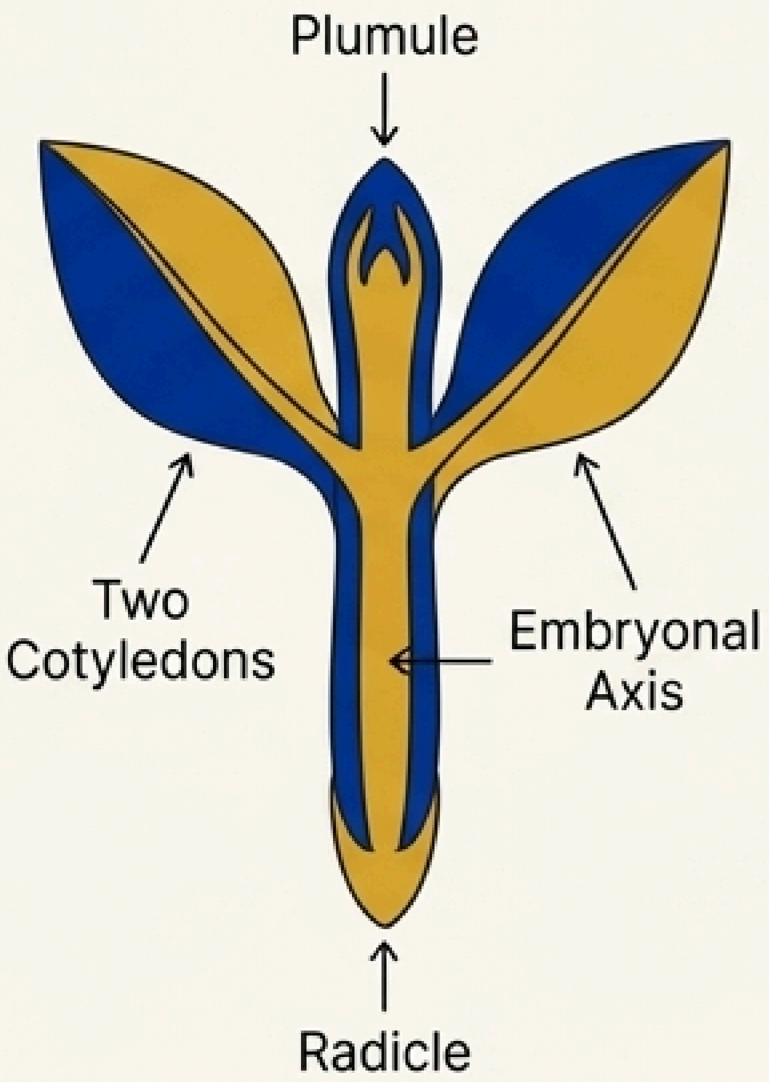
Non-Albuminous (Ex-endospermic) Seeds

Endosperm is completely consumed during development (e.g., Pea, Groundnut).

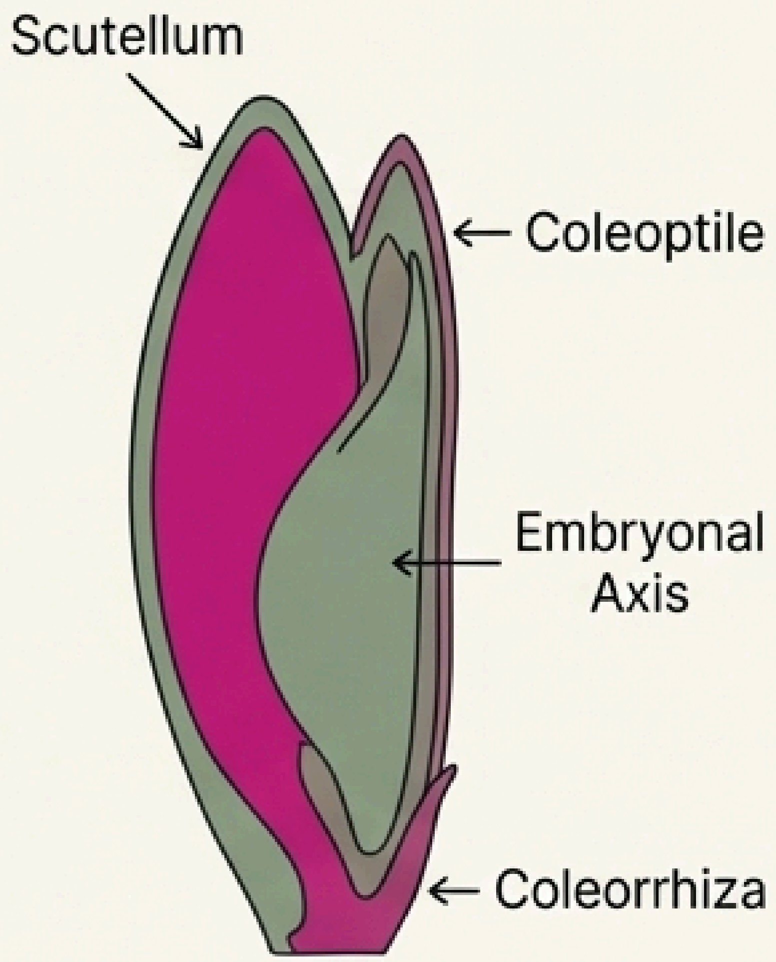


The Embryo

Dicot Embryo



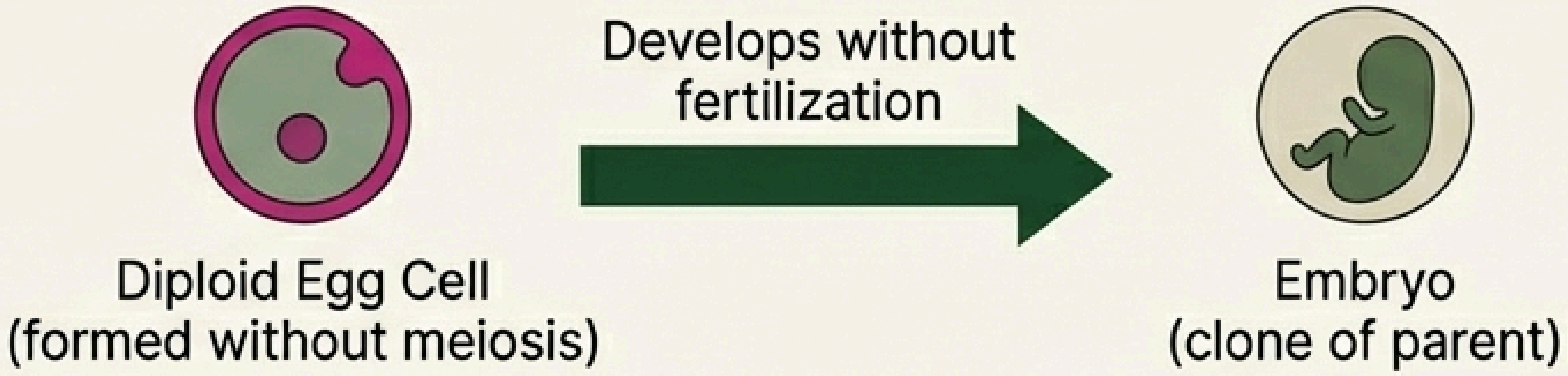
Monocot Embryo



Beyond the Norms: Apomixis and Polyembryony

Apomixis

A special mechanism in some species (e.g., Asteraceae, grasses) to produce seeds **without fertilization.** It is a form of asexual reproduction that mimics sexual reproduction.



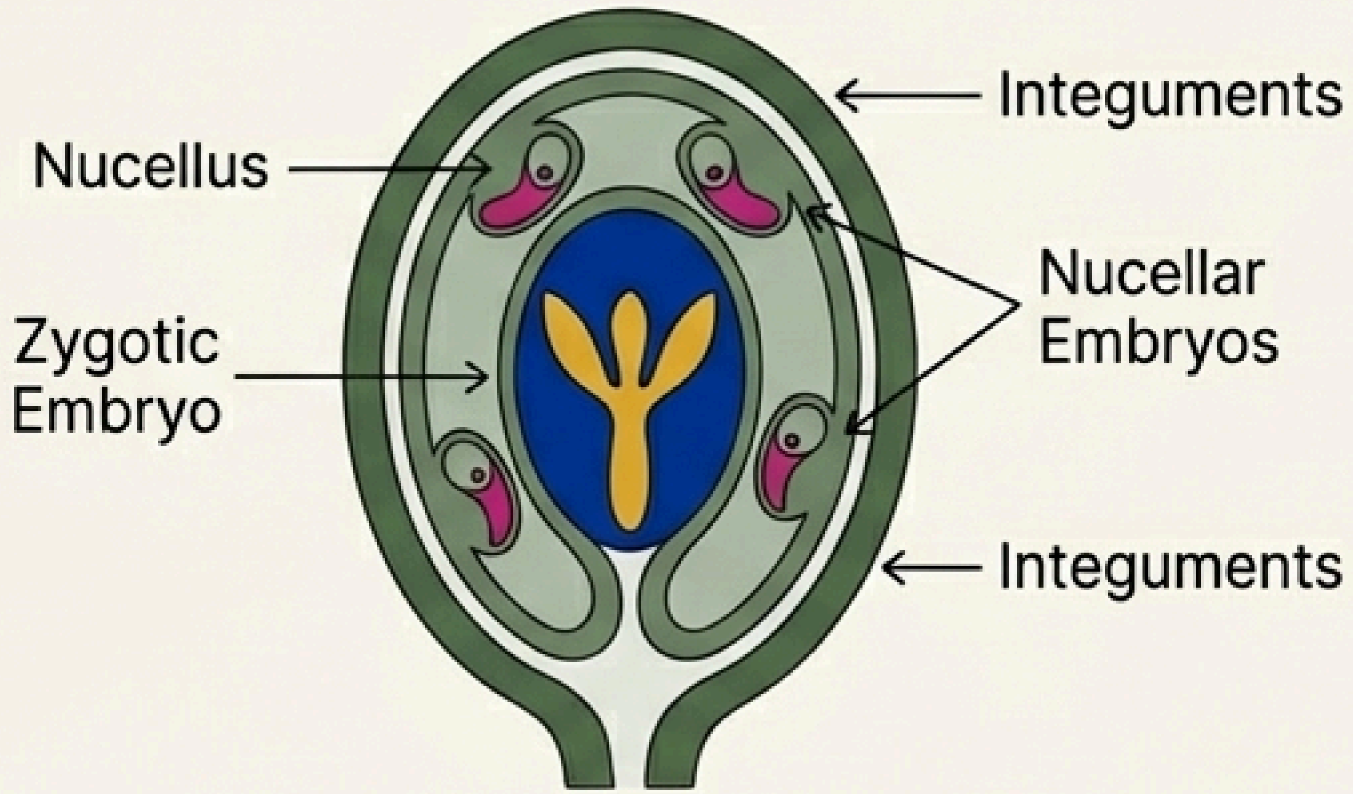
Importance: Allows for the production of “hybrid seeds” where progeny do not segregate characters.

Polyembryony

The occurrence of more than one embryo in a single seed.

Common Examples: Citrus, Mango.

Often, some of the **nucellar cells** surrounding the embryo sac start dividing, protrude into the embryo sac, and develop into additional embryos.



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The Journey in Review: Key Concepts for NEET

Flower (Androecium & Gynoecium)

↓

Microsporogenesis → Pollen Grain

Megasporogenesis → Embryo Sac

Gametogenesis

↓

Pollination (Transfer of pollen to stigma)

↓

Pollen-Pistil Interaction (Pollen tube growth)

↓

n+n=2n n+n+n=3n

Double Fertilization
(Syngamy → Zygote (2n); Triple Fusion → PEN (3n))

↓

Post-Fertilization Development
(Zygote → Embryo; PEN → Endosperm)

↓

Final Products
(Ovule → Seed; Ovary → Fruit)

Understanding this sequence of events—from the differentiation of gametes to the formation of the seed—is fundamental to mastering plant biology.